SOFTWARE TESTING ASSIGNMENT

MODULE 1 – FUNDAMENTAL

1. What is SDLC?

- SDLC (Software Development Life Cycle) is a process used to develop software step-by-step. It includes planning, designing, coding, testing, and maintaining the software.

2. What is Software Testing?

- Software Testing is the process of checking software to find errors or bugs and make sure it works correctly.

3. What is Agile Methodology?

- Agile Methodology is a way to develop software in small parts with continuous testing and feedback.
  It focuses on teamwork, flexibility, and quick delivery.

4. What is SRS?

- SRS (Software Requirement Specification) is a document that explains what the software should do and its requirements.

5. What is OOPs?

- OOPs (Object-Oriented Programming) is a programming method based on objects and classes.
- It helps organize code using concepts like inheritance, encapsulation, polymorphism, and abstraction.

6. Write Basic Concepts of OOPs

- Basic Concepts of OOPs:

- Class → Blueprint to create objects.
- Object → Instance of a class.
- Encapsulation → Wrapping data and methods together.
- Inheritance → One class gets properties of another class.
- Polymorphism → Same method, different behavior.
- Abstraction → Showing important details, hiding complexity.

7. What is Object?

- Object is an instance of a class.

- It represents a real-world thing and contains data and functions.

8. What is Class?

- Class is a blueprint or template used to create objects. It defines data and functions for objects.

9. What is Encapsulation?

- Encapsulation is the process of combining data and methods into one unit (class) and hiding internal details for security.

10. What is Inheritance?

- Inheritance is a feature where one class gets properties and methods from another class. It helps reuse code.

11. What is Polymorphism?

- Polymorphism means one method can have different forms or behaviors. Same function name, different actions.

12. Write SDLC Phases with Basic Introduction

1. Planning Phase:

Define project goals, scope, resources, timeline, and budget. Conduct feasibility study to determine if the project is viable.

2. Analysis Phase:

Gather detailed requirements from stakeholders. Create SRS document. Identify what the system should do.

3. Design Phase:

Create system architecture, database design, UI/UX design. Plan how the system will work technically.

4. Implementation Phase:

Write actual code based on design documents. Developers build the software using programming languages.

5. Testing Phase:

Test the software for bugs and errors. Ensure it meets requirements and works correctly.

6. Deployment Phase:

Release the software to users. Install and configure the system in the production environment.

7. Maintenance Phase:

Provide ongoing support, fix bugs, add new features, and update the software as needed.

13. Explain Phases of the Waterfall Model

- The Waterfall Model is a linear and sequential approach where each phase must be completed before moving to the next. Like a waterfall, progress flows in one direction.

Phases:

1. Requirement Analysis:

Gather all requirements. Create detailed documentation of what needs to be built.

2. System Design:

Design system architecture, database, and interfaces based on requirements.

3. Implementation:

Write code and develop software modules.

4. Testing:

Test the complete system to find and fix defects.

5. Deployment:

Deploy the software to production environment.

6. Maintenance:

Support users and fix issues after deployment.

14. Write Phases of Spiral Model

- The Spiral Model combines iterative development with systematic aspects of the waterfall model. It focuses on risk analysis and allows multiple iterations.

- Four Main Phases (repeated in each spiral):

1. Planning (Identification):

Gather requirements, identify objectives, and define project scope.

2. Risk Analysis:

Identify potential risks and find solutions. Build prototypes to test feasibility.

3. Engineering (Development):

Design, code, and test the software based on risk analysis results.

4. Evaluation (Customer Review):

Get customer feedback and plan for the next iteration.

15. Write Agile Manifesto Principles

- Agile Manifesto Principles:

- Customer satisfaction through early delivery.
- Welcome changes in requirements.
- Deliver working software frequently.
- Teamwork between developers and customers.
- Simple design and continuous improvement.
- Face-to-face communication is best.
- Working software is main progress measure.
- Regular review and improvement.

16. Explain Working Methodology of Agile Model and Pros & Cons

- Working Methodology:

1. Sprint Planning:

Team selects work items from backlog for the upcoming sprint (2-4 weeks).

2. Daily Stand-up:

Short daily meeting where team shares progress and challenges.

3. Development:

Team works on tasks, tests features, and collaborates continuously.

4. Sprint Review:

Demonstrate completed work to stakeholders and get feedback.

5. Sprint Retrospective:

Team reflects on what went well and what can be improved.

Pros (Advantages):

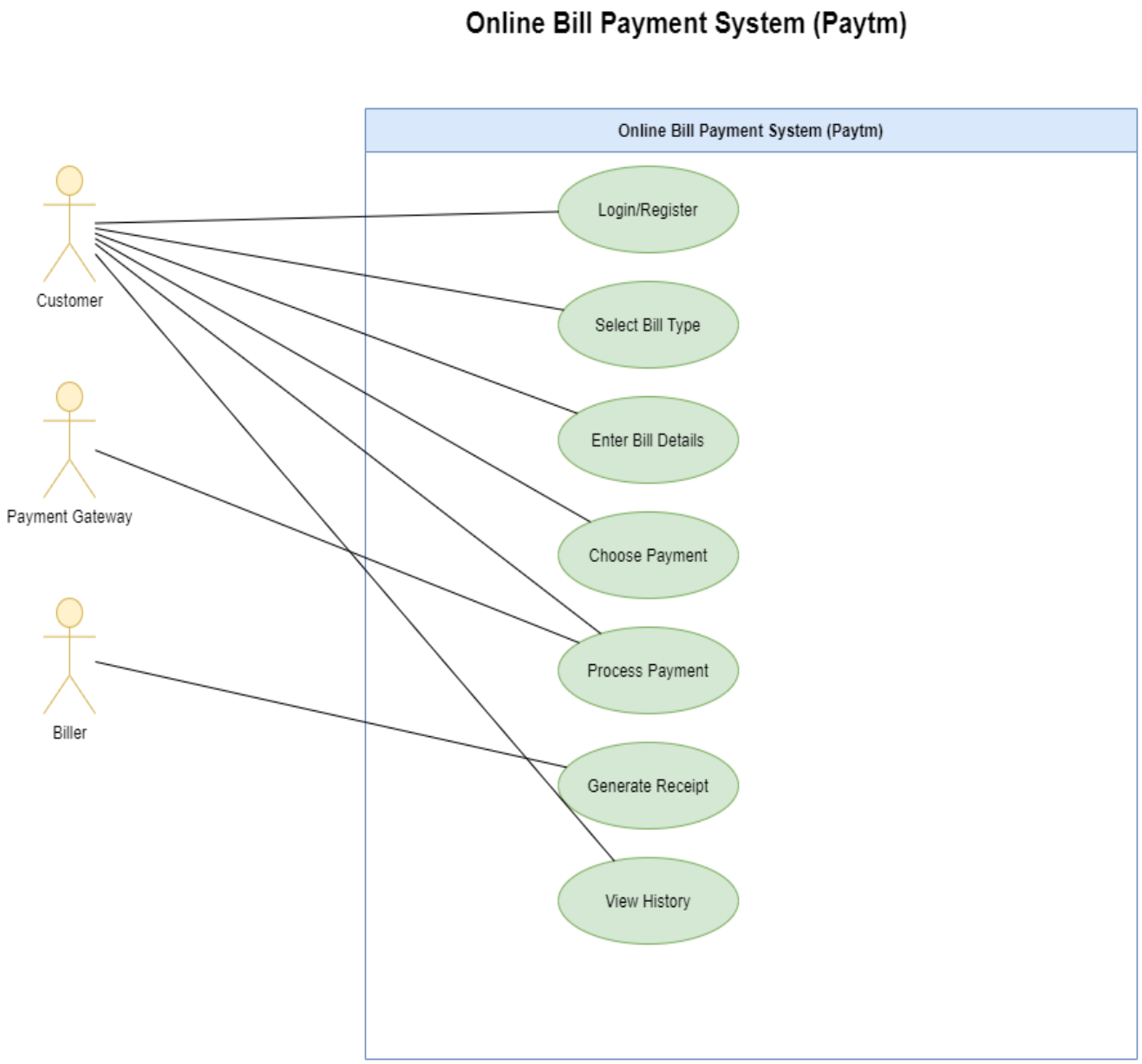
- Faster delivery of working software
- Easy to accommodate changes
- Better collaboration between team and customer
- Continuous feedback and improvement
- Higher customer satisfaction
- Early detection of issues

Cons (Disadvantages):

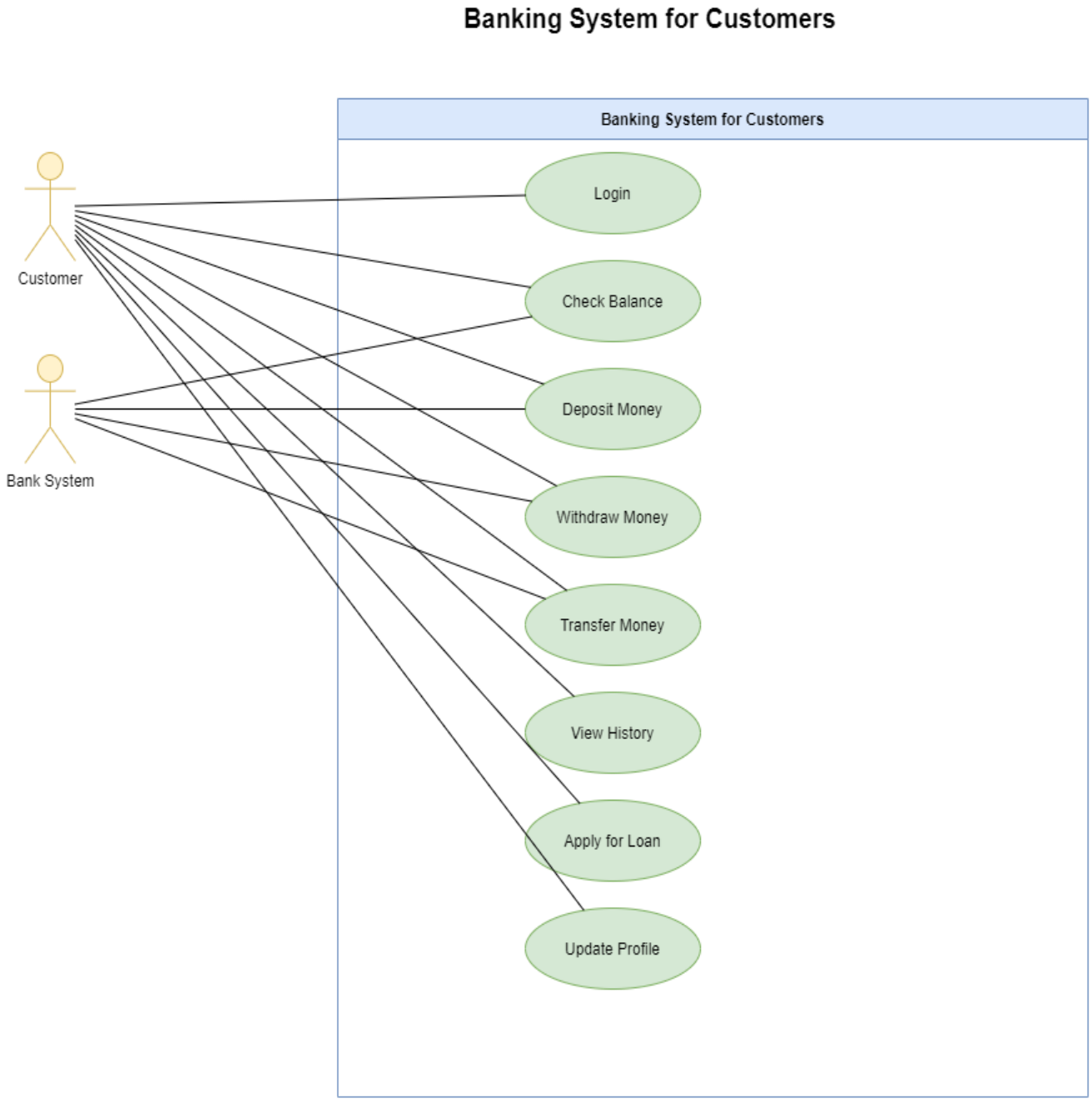
- Requires experienced team members
- Less predictable timelines and costs
- Documentation can be neglected
- Requires active customer involvement
- Not suitable for projects with fixed scope

Use Case Diagrams :-

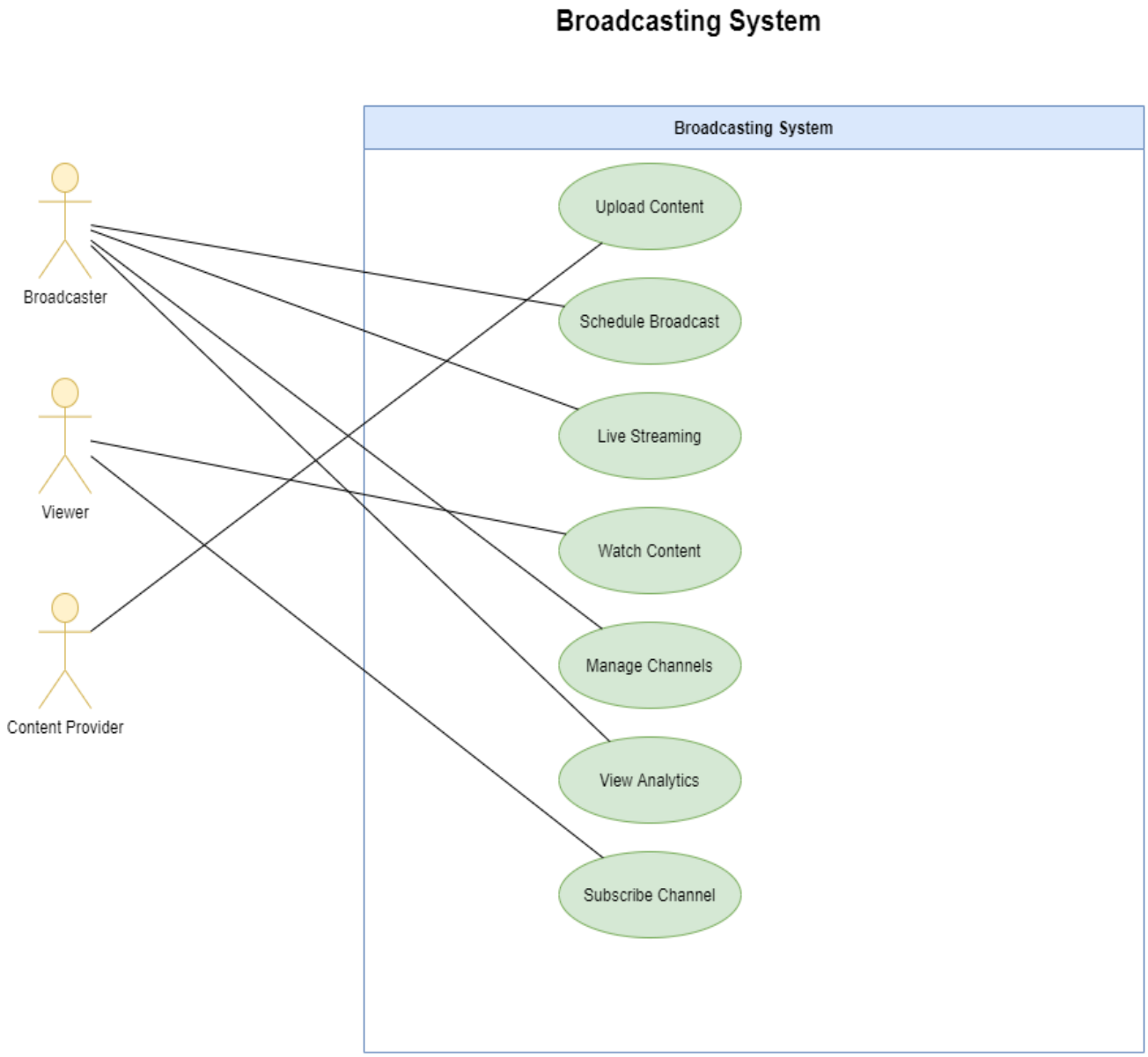
17. Draw Use Case Diagram - Online Bill Payment System (Paytm)



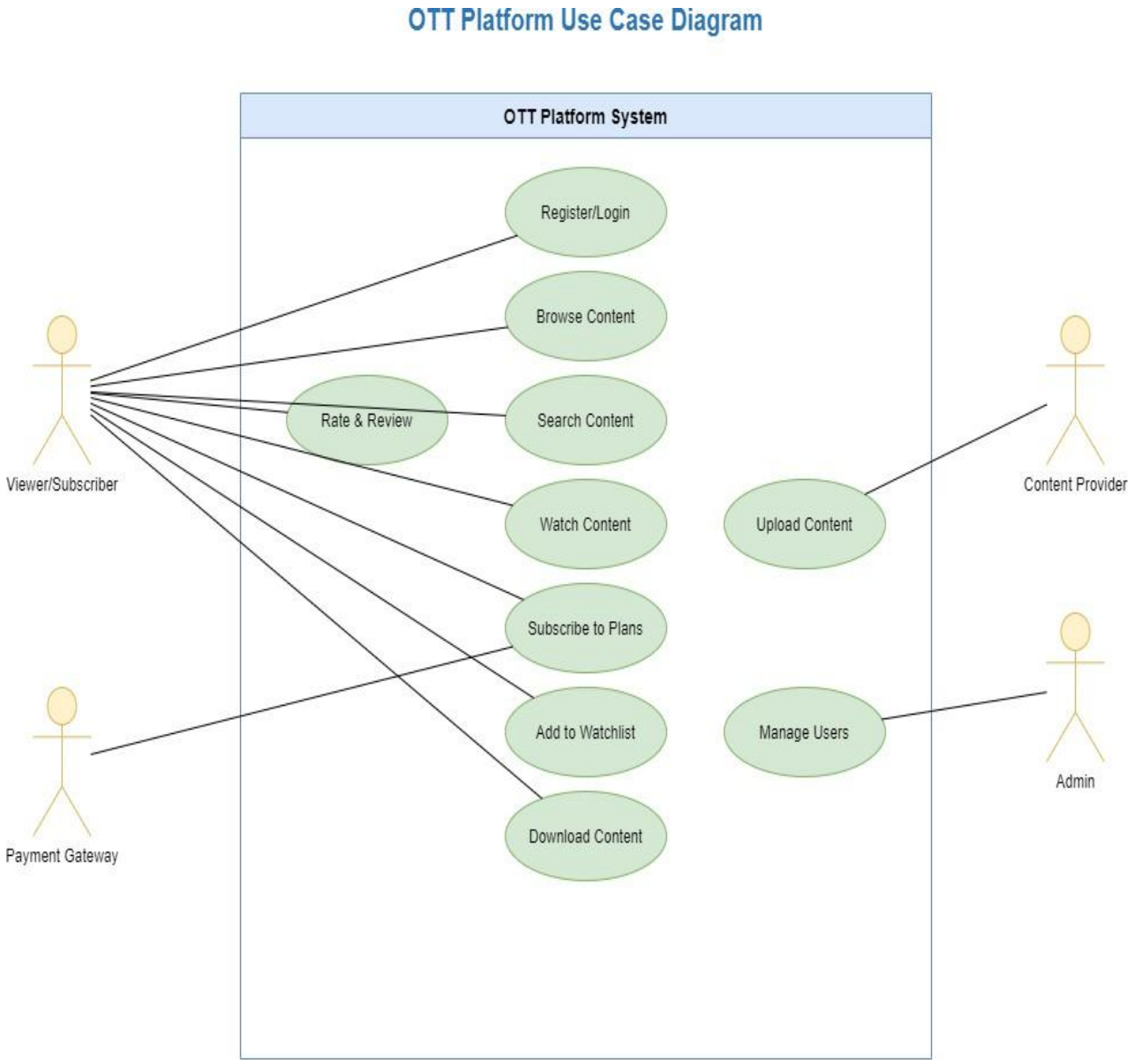
18. Draw Use Case Diagram - Banking System for Customers



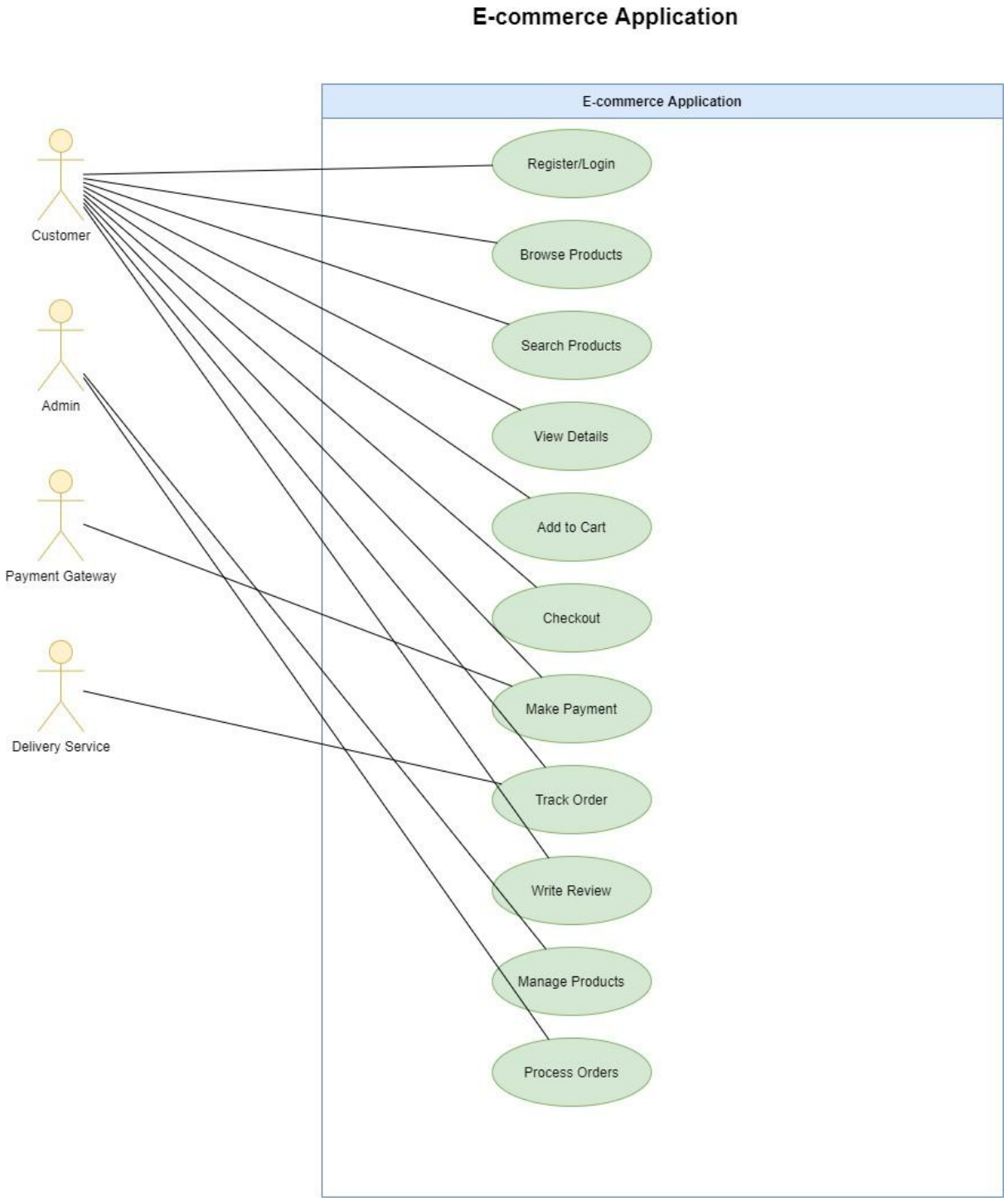
19. Draw Use Case Diagram - Broadcasting System

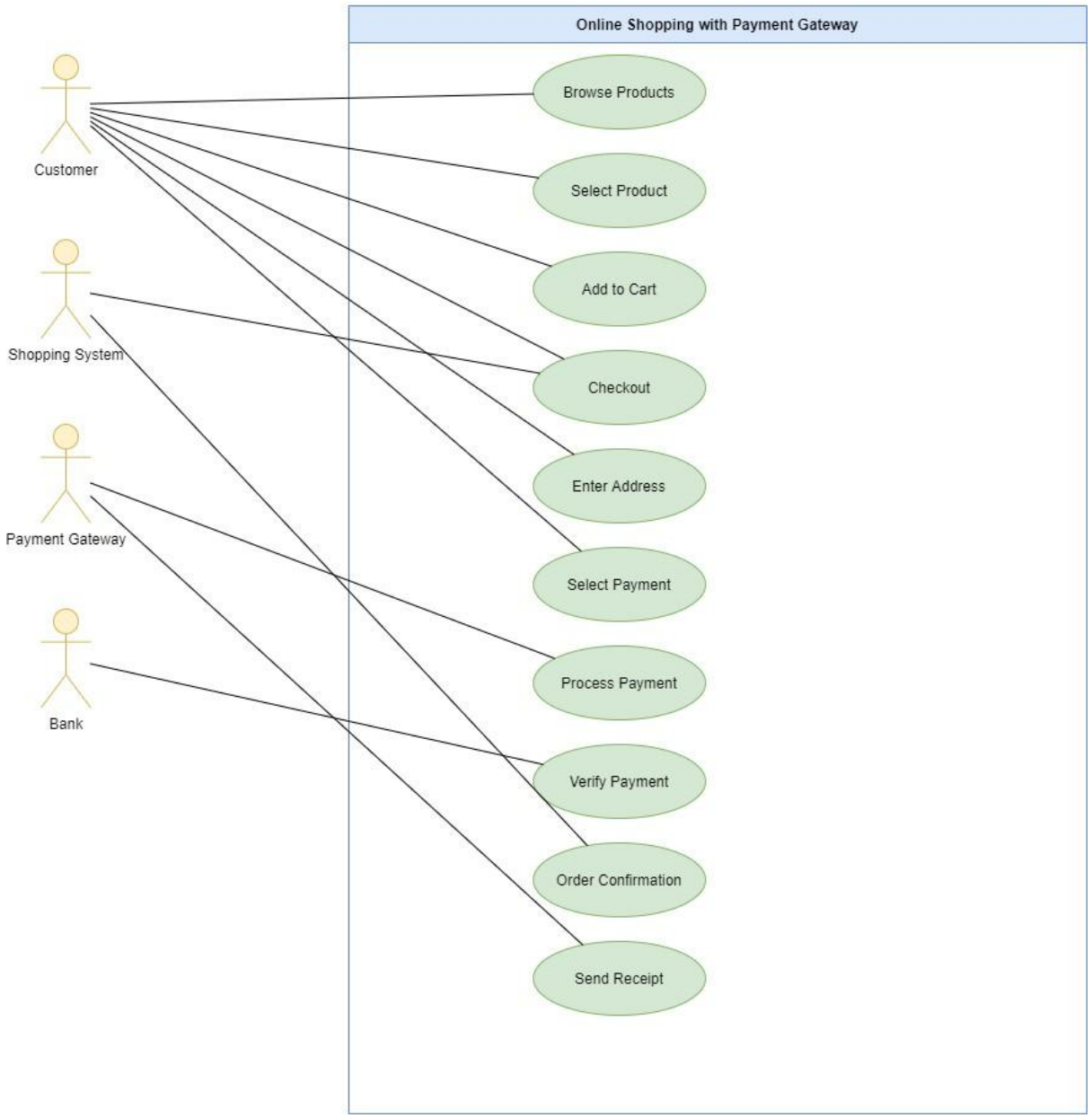


20. Use Case Diagram - OTT Platform



21. Draw Use Case Diagram - E-commerce Application



22. Draw Use Case Diagram - Online Shopping Product Using Payment Gateway

Online Shopping with Payment Gateway